

UPO Golden Gate Assembly + Heat-Shock Transformation

A beginner-friendly tutorial for running W-O-CL-01 on the Opentrons OT-2 · Workflow ID: W-O-CL-01 · Script version: v4.0 · Date: 2026-06-22

This tutorial walks through running **W-O-CL-01_UPO_GGA_heatshock.py**: a Golden Gate assembly of a standard pPIC backbone with 20 UPO gBlocks plus a no-insert negative control, followed by competent-cell mixing and heat-shock transformation. As of v4.0, the robot handles almost everything by itself — backbone, master mix, all 20 gBlocks, and the negative control are pipetted automatically. You're only needed twice: to load fresh competent cells partway through, and to plate the finished transformations at the end.

New to the OT-2? This guide assumes no prior experience. If you've run protocols before, feel free to skip straight to “Step-by-step walkthrough.”

1. OT-2 basics, in brief

The OT-2 is a liquid-handling robot. It runs a **protocol** — a Python script that tells it exactly which labware to expect in which deck position, and what to do with it. You load the protocol file into the **Opentrons App** on a connected computer, the app checks it against the robot, and you press **Run**.

- **Deck slots:** the OT-2's work surface is divided into 11 numbered slots (1–11) plus a fixed trash bin (slot 12). Each slot holds one piece of labware or one module.
- **Modules:** powered attachments that clip into a deck slot and do something beyond basic pipetting. This protocol uses one: the **Thermocycler**.
- **Single-nozzle mode:** normally, the OT-2's 8-channel pipettes move all 8 tips together. For several steps in this protocol, the P20 is reconfigured to use just *one* of its 8 nozzles — letting it act like a single-channel pipette and reach into one specific tube at a time. This is how the robot can pick up an individual gBlock tube without disturbing its neighbours. You don't need to do anything for this — it's built into the script — but you'll see it mentioned in the steps below.
- **Pauses:** when a protocol needs you to do something by hand, it calls a *pause*. The robot stops completely and the Opentrons App shows the message you need to act on. Nothing proceeds until you click **Resume** — there's no time pressure, take the time you need.
- **Don't touch the deck while it's running** (only at a pause). If you need to abort, use the app's Cancel button rather than physically interfering with the robot.

2. What you'll need

Hardware

Item	Role in this protocol
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OT-2 robot	Runs the whole show
P20 8-channel pipette (GEN2)	Backbone, master mix, water, gBlocks, negative control
P300 8-channel pipette (GEN2)	Competent cells (50 uL/well)
Thermocycler module	Golden Gate cycling; heat-shock + recovery
1x tip rack, 20 uL	For the P20
1x tip rack, 300 uL	For the P300
1x 24-tube rack (1.5/2 mL tubes)	Backbone (1 tube), master mix (1 tube), 20 individual gBlock tubes
1x 12-well reservoir (e.g. NEST 15 mL)	Water and competent cells
1x Bio-Rad 96-well 200 uL PCR plate	The reaction plate, stays on the thermocycler the whole run

Reagents (prepare before you start)

Reagent	Per well	Minimum needed (21 wells)	Suggested to prepare
Backbone (normalized stock)	1 uL	21 uL	~30-50 uL, in its own tube
2X Golden Gate master mix	10 uL	210 uL	~250 uL, in its own tube
Water	8 uL	168 uL	~250 uL, in the reservoir
Competent cells (on ice until needed)	50 uL	1050 uL	~1.2 mL, in the reservoir

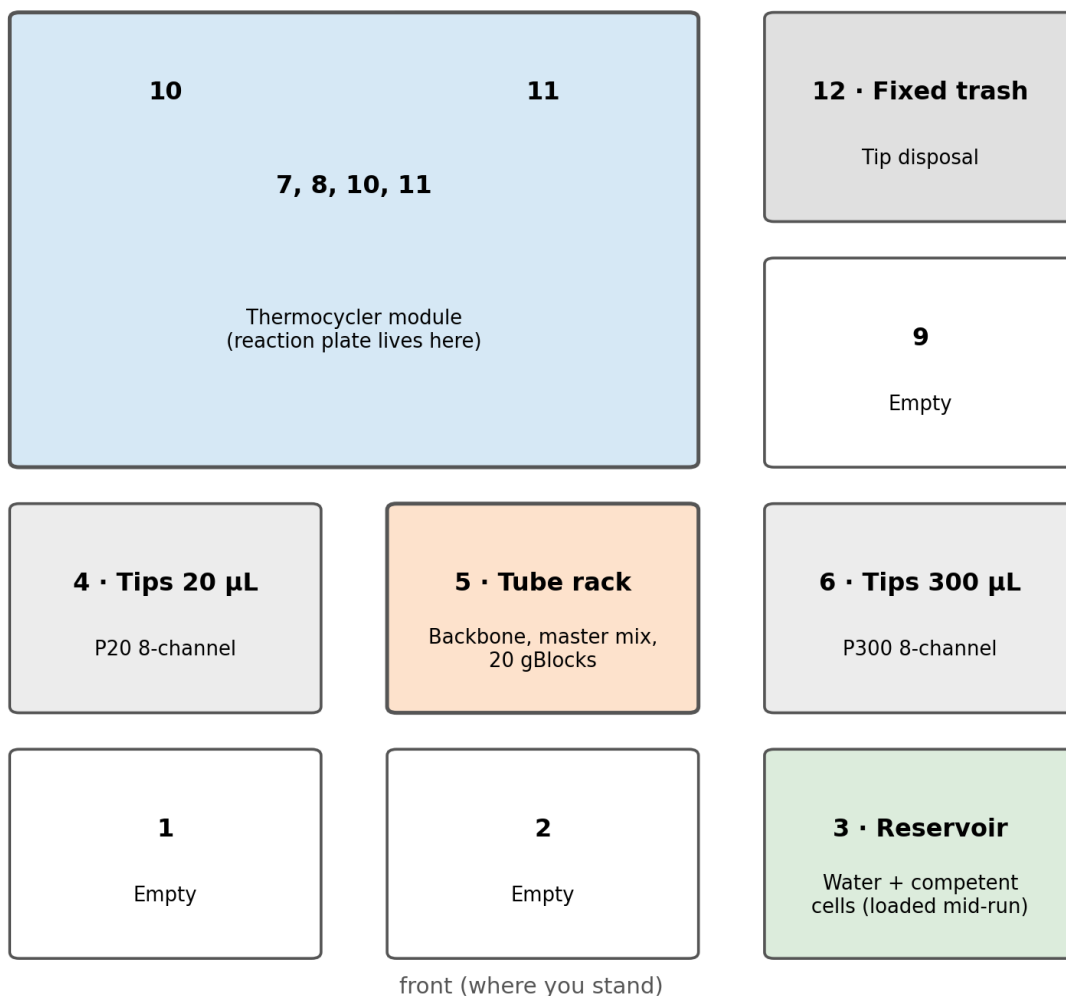
Note: 21 wells are prepped (the 20 real reactions plus 1 negative control). Backbone and master mix each need just *one* tube — not a full row of wells — since the robot now picks them up with a single pipette nozzle. The "minimum needed" volumes don't include pipetting dead volume, hence the larger suggested-prep amounts. Competent cells are the one reagent loaded *mid-run*, not at the start — see Step U-M-01.

Also have ready: your 20 UPO gBlocks, each resuspended in its own tube with at least a few uL (the protocol only draws 1 uL from each, but normal handling volume is fine), and individual agar plates for the final manual plating step.

3. Deck layout

Set up the deck exactly as shown below before starting the run. The tube rack holds backbone, master mix, and all 20 gBlocks — see the loading map in Section 6 for exactly which tube goes where.

OT-2 deck — top view (back of robot ↑)



Slot	Contents	Notes
1	Empty	—
2	Empty	—
3	12-well reservoir	Water (A1), competent cells (A2, loaded mid-run)
4	20 uL tip rack	For the P20
5	24-tube rack	Backbone (A6), master mix (B6), 20 gBlocks (columns 1-5)
6	300 uL tip rack	For the P300
7, 8, 10, 11	Thermocycler module	Reaction plate stays here for the entire run

9	Empty	—
12	Fixed trash	Always present, tip disposal

4. Modules used in this protocol

Thermocycler

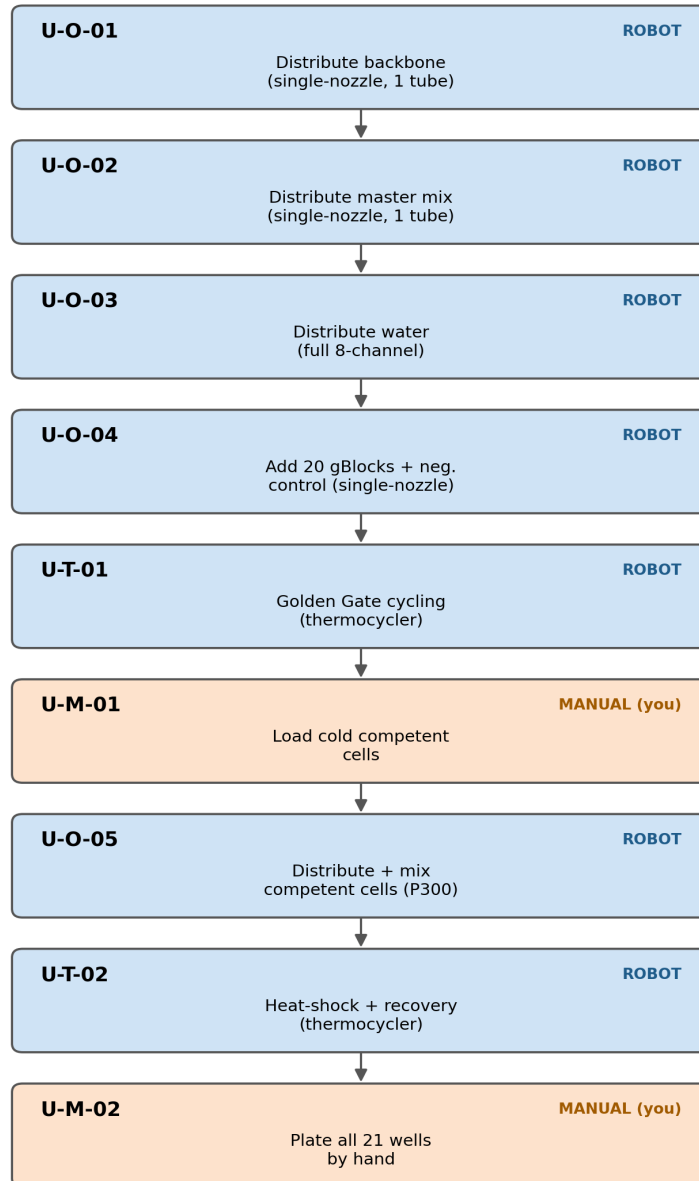
Runs every precise temperature step: the Golden Gate digestion/ligation cycling, the enzyme deactivation step, and later the heat-shock pulse and recovery incubation. The reaction plate lives here for the entire run — it never has to move.

No Heater-Shaker in this version. An earlier design used one to mix competent cells in gently, but plain pipette mixing (with a slowed-down flow rate) turned out to be the more robust choice — it's what almost every published transformation protocol actually does, and it avoids two manual plate transfers and a permanently unusable deck slot. The lid temperature for both thermocycler profiles below is also tuned to each profile's own maximum sample temperature, rather than reusing one PCR-style setting for everything.

5. Step-by-step walkthrough

Nine steps in total. The robot runs seven of them unattended; you're needed for the other two.

Robot does this automatically You do this by hand



U-O-01 — Distribute backbone**ROBOT — hands off**

The P20, in single-nozzle mode, picks up one tip and repeatedly aspirates from the single backbone tube (well A6 of the tube rack), dispensing 1 uL into each of the 21 reaction wells. It re-aspirates whenever the tip runs low, all with the same tip throughout — safe to reuse since it's the same reagent every time.

U-O-02 — Distribute master mix**ROBOT — hands off**

Same pattern as backbone, sourcing 10 uL per well from the master mix tube (well B6).

U-O-03 — Distribute water**ROBOT — hands off**

The P20 switches back to normal full 8-channel mode for this one step, since water is sourced from a true reservoir channel — exactly what full 8-channel access is efficient and safe for. 8 uL goes into every well.

U-O-04 — Add the 20 gBlocks + negative control**ROBOT — hands off**

Back to single-nozzle mode. For each of the 20 gBlocks, the robot picks up a fresh tip, draws 1 uL from that gBlock's own tube, and dispenses it into its mapped reaction well (see the tube and well maps in Section 6). A fresh tip each time, since these are 20 different constructs and cross-contamination between them would compromise the comparison this experiment exists to make. As a final iteration of the same loop, 1 uL of plain water (not gBlock) goes into the negative control well — this lets you tell “construct doesn't work” apart from “backbone never got an insert.”

U-T-01 — Golden Gate cycling**ROBOT — hands off**

The thermocycler closes its lid and cycles between 37°C and 16°C 25 times, then holds at 60°C for 5 minutes to deactivate the enzyme, then cools to 4°C and opens the lid. This takes roughly an hour — a good point to step away.

U-M-01 — Load cold competent cells**MANUAL — your turn**

The only reagent-loading pause left. Place freshly-thawed, on-ice competent cells into reservoir well A2 — loading them now rather than at the start keeps them cold for as little time as possible before use. Click Resume once done.

U-O-05 — Distribute + mix competent cells**ROBOT — hands off**

The P300 (full 8-channel, as normal) adds 50 uL of competent cells to every well, then gently mixes 3 times with a slowed-down flow rate — competent cells are shear-sensitive, so this is deliberately gentler than the pipette's default speed.

U-T-02 — Heat-shock + recovery**ROBOT — hands off**

The thermocycler runs the full heat-shock profile: 4°C cold soak (10 min) → 42°C pulse (90 sec) → 4°C (2 min) → 37°C recovery hold (1 hour). It then opens the lid and deactivates.

Final pause. Plate each of the 20 transformations plus the negative control onto its own agar plate — see Section 7 for exactly which well is which. The negative control should show few or no colonies; treat results from the real wells with extra caution if it doesn't. Nothing in this script does the plating itself — it's all yours from here.

6. Tube rack & well maps (reference)

Load the tube rack like this before starting the run:

Row	Col 1	Col 2	Col 3	Col 4	Col 5	Col 6
A	UPO_01	UPO_05	UPO_09	UPO_13	UPO_17	Backbone
B	UPO_02	UPO_06	UPO_10	UPO_14	UPO_18	Mastermix
C	UPO_03	UPO_07	UPO_11	UPO_15	UPO_19	— spare
D	UPO_04	UPO_08	UPO_12	UPO_16	UPO_20	— spare

Once the robot finishes U-O-04, here's where everything ends up on the reaction plate:

Row	Column 1	Column 2	Column 3
A	UPO_01	UPO_09	UPO_17
B	UPO_02	UPO_10	UPO_18
C	UPO_03	UPO_11	UPO_19
D	UPO_04	UPO_12	UPO_20
E	UPO_05	UPO_13	Negative control
F	UPO_06	UPO_14	— (unused)
G	UPO_07	UPO_15	— (unused)
H	UPO_08	UPO_16	— (unused)

Row H of column 3 is the only truly unused position: 20 gBlocks + 1 negative control = 21 of the 24 available wells.

7. Notes & troubleshooting

- **"How can a multichannel pipette pick up just one gBlock tube?"** Normally an 8-channel pipette moves all 8 tips together, and that really can't reach into one arbitrary tube safely. But in single-nozzle mode, only one nozzle actually carries a tip — the other seven are bare, shorter nozzle bodies that stay well clear of neighbouring tube caps. That's a real, documented OT-2 feature (API 2.20+), not a workaround.

- **"Why does backbone/master mix distribution take a while?"** Each only has a 20 uL max per aspirate, so for master mix especially (10 uL/well) the tip refills frequently. This trades a bit of robot time for not needing to pre-fill 8 wells of a plate with the same reagent — worth it given backbone in particular can be precious.
- **"Mixing looks uneven after competent cells go in."** The mix step already runs slower than default specifically to be gentle on the cells. If it still looks streaky, that's more likely a viscosity/volume issue than a speed one — don't compensate by increasing flow rate, since competent cells are shear-sensitive.
- **"Our master mix isn't a premade 2X kit."** The 10 uL master-mix volume assumes a single premixed 2X Golden Gate enzyme+buffer reagent. If your kit supplies enzyme and buffer separately, edit the PARAMETERS block at the top of the script and add a second tube + distribution step rather than changing any of the downstream logic.
- **"Our backbone isn't a flat 1 uL per well."** The script assumes a normalized backbone stock where a constant 1 uL works for every well. If you're working from molarity instead, adjust VOL_BACKBONE_UL (and re-balance VOL_WATER_UL so the total stays at 20 uL).
- **Is the backbone actually Golden-Gate-ready?** This script assumes your pPIC backbone already has Bsal sites flanking the insertion site. If that domestication hasn't been done yet, that's a separate cloning step that needs to happen before this protocol is run.
- **Changing the number of UPOs?** Edit NUM_UPOS at the top of the script — the reaction columns, tube map, well map, and negative control position all recompute automatically. The tube rack holds up to 22 gBlocks alongside backbone and master mix; beyond that, a second tube rack would be needed.

Parameters quick reference

All of the values below live in one block near the top of the script — that's the only place you should need to edit.

Parameter	Value	Meaning
VOL_BACKBONE_UL	1.0	Backbone per well
VOL_MASTER_MIX_UL	10.0	2X GG master mix per well
VOL_WATER_UL	8.0	Water per well
VOL_GBLOCK_UL	1.0	gBlock per well
VOL_COMPETENT_CELLS_UL	50.0	Competent cells per well
MIX_FLOW_RATE_UL_PER_S	40.0	Slowed flow rate during cell mixing
GG_CYCLE_REPETITIONS	25	Golden Gate digestion/ligation cycles
GG_LID_TEMP_C	70	Thermocycler lid temp during GG cycling
HS_LID_TEMP_C	47	Thermocycler lid temp during heat-shock
HS_RECOVERY_TIME_S	3600	Post-heat-shock recovery hold (1 h)